Forecasting US Stock Closing Dynamics

Final Project

Naël Arnoux
École CentraleSupélec
Paris, France
nael.arnoux@student-cs.fr

Marta Shkreli
École CentraleSupélec
Paris, France
marta.shkreli@student-cs.fr

Wei Luo
École CentraleSupélec
Paris, France
wei.luo@student-cs.fr

***Abstract*—This project investigates the dynamics of the Nasdaq Closing Cross auction using high-frequency order book and auction data from the Optiver Trading at the Close dataset. Final prices are established and used as official market close values during the closing auction, which is a very volatile and information-dense part of the trading day. This study aims to model and forecast short-term price movements in the last minutes of trading by utilising machine learning techniques.**

***Index Terms*—Stock price forecasting, time series analysis, machine learning, financial markets, predictive modeling**

I. Project Overview

This project focuses on building a machine learning framework to analyze and forecast short-term price movements during the Nasdaq Closing Cross auction. Using historical high-frequency auction and order book data, the study explores how order imbalance, bid-ask dynamics, and temporal proximity to the market close influence price changes. The project emphasizes both predictive performance and interpretability. Our pipeline includes large-scale preprocessing, auction microstructure feature engineering (spread, imbalance pressure, price reference divergences), and regression/time-series formulations designed to capture both cross-sectional and temporal dynamics of closing-price formation.

II. Problem Adressed

The closing auction is one of the most critical but least predictable moments in the trading day. Prices formed during this phase affect portfolio valuation, index calculation, and trading strategies across global markets. This project addresses the difficulty of predicting near-term price movements during the closing auction using high-frequency market data, while accounting for market microstructure effects and avoiding information leakage inherent in temporal data.

A. Research Questions

- Can historical price patterns be used to predict future closing prices?
- Which models best capture short-term and long-term market dynamics?
- What features contribute most to predictive performance?

III. Potential Applications

The findings and models developed in this project can be applied in several domains:

- *Algorithmic Trading*: Enhancing execution strategies around the market close to reduce price impact and slippage.
- *Market Microstructure Analysis*: Providing empirical insights into how order imbalance and liquidity shape price formation.
- *Risk Management*: Improving short-horizon risk estimation during volatile market phases.
- *Financial Research*: Supporting further studies on auction-based price discovery and high-frequency market behavior.

IV. Problem Definition

A. Problem Description

Given high-frequency snapshots of order book prices, volumes, and auction imbalance indicators, the objective is to predict the magnitude of near-term price movements during the closing phase. The main challenge lies in extracting informative signals from noisy, rapidly evolving data while ensuring robustness across stocks and trading days.

B. Dataset and Columns

The dataset used in this project is sourced from the Optiver *Trading at the Close* Kaggle competition and consists of high-frequency auction and order book observations for multiple Nasdaq-listed stocks.

Key columns include:

- `stock_id` — security identifier
- `date_id` — trading day identifier
- `time_id` — intraday time index
- `seconds_in_bucket` — seconds remaining / position within the auction time bucket
- `imbalance_size` — size of auction imbalance
- `imbalance_buy_sell_flag` — imbalance direction (buy vs sell)
- `matched_size` — currently matchable size in auction
- `reference_price` — reference price used by the auction mechanism
- `far_price` — indicative far price

- `near_price` — indicative near price
- `bid_price`, `bid_size` — best bid price and size
- `ask_price`, `ask_size` — best ask price and size
- `wap` — weighted average price
- `row_id` — unique row identifier
- **Train only:** `target` — short-horizon price movement to predict
- **Test only:** `currently_scored` — whether the row is scored in evaluation

C. Optimization Objectives

1) Regression Objective: The regression objective is to predict the magnitude of short-horizon price movement during the closing auction using auction and order book features. Model performance is evaluated using Mean Absolute Error (MAE) as the primary metric, consistent with the competition evaluation criterion.

2) Time-Series Objective: The time-series objective is to capture short-term temporal dependencies in auction dynamics by incorporating lag-based features that describe how imbalance and price-related signals evolve over time within the closing window.

V. Methodology

A. Ridge Regression

Ridge Regression is a regularized linear regression model that penalizes large coefficient values. It extends ordinary linear regression by discouraging the model from relying excessively on any single feature, thereby improving numerical stability and robustness. By shrinking coefficients toward zero, Ridge Regression helps reduce overfitting, especially in settings where predictors exhibit strong correlations.

Ordinary linear regression estimates the coefficient vector by minimizing the residual sum of squares:

$$\hat{\beta} = \arg\min_{\beta} \sum_{i=1}^{n} \left(y_i - \mathbf{x}_i^\top \beta\right)^2 .$$

Ridge Regression modifies this objective by introducing an L2 regularization term:

$$\hat{\beta}^{\text{ridge}} = \arg\min_{\beta} \left[\sum_{i=1}^{n} \left(y_i - \mathbf{x}_i^\top \beta\right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right],$$

where $\lambda \geq 0$ controls the strength of the regularization. Larger values of λ impose stronger penalties on large coefficients, leading to more stable and less variance-prone parameter estimates.

In this project, Ridge Regression serves as a stable linear baseline for modeling short-term price movements. This choice is particularly appropriate given the strong multicollinearity among order book price and volume features, where regularization mitigates coefficient instability while preserving interpretability.

1) Feature Engineering for Ridge Regression: Ridge regression is a linear model that captures only linear relationships between predictors and the target. It is not suitable for complex nonlinear patterns and large numbers of feature interactions.

Following the data cleaning, this dataset contains consistent and reliable market information throughout the time interval. Although the original variables describe price and volume, they do not explicitly reflect trading pressure or liquidity. To address this limitation, we construct two engineered features that enhance the model's ability to capture market microstructure dynamics.

a) Directional Imbalance Pressure.: The directional imbalance pressure is defined as:

$$\text{ImbalancePressure}_t = \text{ImbalanceSize}_t \times \text{ImbalanceFlag}_t,$$

where ImbalanceSize_t represents the magnitude of order imbalance at time t, and $\text{ImbalanceFlag}_t \in \{-1, +1\}$ indicates the selling or buying direction. This feature combines both the strength and direction of order imbalance into a single signed variable. Positive values correspond to net buying pressure, while negative values indicate net selling pressure. This feature is included in Ridge regression because it is a simple linear transformation of existing variables with a clear economic interpretation. By aggregating magnitude and direction into a single variable, imbalance pressure provides a concise representation of trading pressure while remaining compatible with the linear assumptions of the model.

b) Bid–Ask Spread.: The bid–ask spread is computed as:

$$\text{Spread}_t = \text{AskPrice}_t - \text{BidPrice}_t.$$

The spread serves as a proxy for market tightness and liquidity conditions. A larger spread typically reflects higher uncertainty or lower liquidity in the market. The bid–ask spread is added to Ridge regression as it captures liquidity conditions through a linear combination of price variables. As a widely used microstructure indicator, the spread enhances interpretability without introducing nonlinear interactions, making it well suited for a regularized linear model.

Together, these engineered features enrich the linear model by making trading pressure and liquidity conditions explicit, while preserving the simplicity and stability required by Ridge regression.

c) Data Usage for Modeling: For the Ridge regression model, we select the following features:

- **Core price information:** `bid_price`, `ask_price`, `wap`, `reference_price`
- **Volume and imbalance:** `matched_size`, `imbalance_size`
- **Engineered features:** `imbalance_pressure`, `spread`

B. Random Forest

Random forest is a non linear ensemble learning method, and it serves as a complement linear baseline models. Basically, it consists of an ensemble of decision trees, where each

tree is trained on a bootstrap sample of the original dataset. The final prediction is obtained by averaging the predictions of all individual trees:

$$\hat{y}(x) = \frac{1}{T}\sum_{t=1}^{T} h_t(x),$$

where $h_t(x)$ denotes the prediction of the t-th decision tree and T is the total number of trees.

Unlike regularized linear regression, random forest does not assume a linear relationship between input features and target variables, making it well suited for high-frequency financial data. Because some feature does not contain a linear relationship, for example, the order of book state and price movement, some features contain a strong interaction actions, such as, price and volume.

1) Feature Engineering for Random Forest: Unlike Ridge regression, which requires simple linear transformations, Random Forest benefits from nonlinear relationships data. We design engineered features to align with the strengths of tree-based models that their ability is to capture complex interactions and decision boundaries. As a result, we emphasize three key aspects of market microstructure: liquidity pressure, market tightness, and short-term dynamics. This approach improves model performance and avoid feature redundancy.

a) imbalance pressure: The imbalance pressure at time t is defined as:

$$\text{Imbalance Pressure}_t = \text{Imbalance Size}_t \times \text{Imbalance Buy Flag}_t$$

where $\text{Imbalance BuyFlag}_t \in \{-1, +1\}$ indicates selling or buying direction.

b) bid–ask: The bid–ask spread at time t is defined as:

$$\text{Spread}_t = \text{Ask Price}_t - \text{Bid Price}_t$$

c) Momentum-Based Features.: To capture short-term dynamics in trading activity, we compute first-order differences of key volume variables within each stock-day group to avoid temporal leakage across different trading sessions:

$$\text{Imbalance Size Diff}_t = \text{Imbalance Size}_t - \text{Imbalance Size}_{t-1}, \tag{1}$$

$$\text{Matched Size Diff}_t = \text{Matched Size}_t - \text{Matched Size}_{t-1}. \tag{2}$$

The imbalance size difference captures acceleration in order imbalance pressure, while the matched size difference reflects sudden bursts in trading activity. For the first observation of each day, these differences are set to zero.

d) Data Usage for Modeling: : For Random Forest model, we select the following feature:

2) Feature Selection for Random Forest: For the Random Forest model, we select the following ten features:

- **Core price features:** `bid_price`, `ask_price`, `wap`, `reference_price`
- **Volume and imbalance:** `matched_size`, `imbalance_size`
- **Engineered nonlinear signals:** `imbalance_pressure`, `spread`
- **Short-term dynamics:** `imbalance_size_diff`, `matched_size_diff`

C. LightGBM Regression Model

Light Gradient Boosting Machine (LightGBM) is a tree-based ensemble learning algorithm based on gradient boosting decision trees. Unlike linear models, LightGBM is capable of capturing complex nonlinear relationships and feature interactions, making it particularly suitable for high-frequency financial data where price dynamics are driven by heterogeneous and non-additive effects.

Given a set of input features x_i and corresponding target values y_i, the model constructs an additive ensemble of regression trees:

$$\hat{y}_i = \sum_{k=1}^{K} f_k(x_i), \quad f_k \in \mathcal{F},$$

where each f_k represents a decision tree in the ensemble and $\mathcal{F}$ denotes the space of regression trees. The model is trained by minimizing a regularized loss function of the form:

$$\mathcal{L} = \sum_{i=1}^{n} \ell(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k),$$

where $\ell(\cdot)$ is the loss function and $\Omega(\cdot)$ is a regularization term that penalizes model complexity. In this project, the loss function is the Mean Absolute Error (MAE), which is robust to extreme price movements commonly observed during the closing auction.

LightGBM employs gradient-based one-side sampling and histogram-based splitting, allowing it to efficiently scale to large datasets while maintaining strong predictive performance.

1) Feature Engineering for LightGBM: The LightGBM model leverages both raw auction variables and engineered features derived from market microstructure principles. In addition to the original order book and auction fields, the following engineered features are constructed:

- **Directional Imbalance Pressure**, defined as:

  $$\text{ImbalancePressure}_t = \text{ImbalanceSize}_t \times \text{ImbalanceBuySellFlag}_t,$$

  which combines the magnitude and direction of order imbalance into a single signed variable representing net buying or selling pressure.
- **Bid–Ask Spread**, computed as:

  $$\text{Spread}_t = \text{AskPrice}_t - \text{BidPrice}_t,$$

  serving as a proxy for instantaneous liquidity and market tightness.
- **Auction Price Divergence**, defined as the difference between indicative far and near prices:

  $$\text{AuctionDivergence}_t = \text{FarPrice}_t - \text{NearPrice}_t,$$

capturing uncertainty in auction price formation.

To incorporate short-term temporal dynamics, lag-based and first-difference features are computed for key variables, including imbalance size, matched size, reference price, and imbalance pressure. These features allow the model to learn how rapid changes in auction conditions influence subsequent price movements.

Additionally, a global imbalance mean feature is introduced to capture market-wide pressure across all stocks at a given time, providing cross-sectional context beyond single-stock information.

Feature Selection: The feature set is selected to jointly capture instantaneous market conditions, short-term temporal dynamics, and market-wide context during the closing auction. Static microstructure variables describe liquidity and imbalance at a given instant, while lag-based difference features encode rapid changes in order flow and price references, and the global imbalance mean provides cross-sectional information on overall market pressure.

2) Model Training and Validation: The LightGBM model is trained using a chronological train–validation split to preserve temporal ordering and prevent information leakage. The training objective is MAE, consistent with the competition evaluation metric and appropriate for handling heavy-tailed price movements.

Early stopping is applied based on validation performance to prevent overfitting and to select the optimal number of boosting iterations. The final model configuration balances model complexity and generalization performance, making it suitable for large-scale, high-frequency auction data.

Model hyperparameters: Model hyperparameters are chosen to balance expressive power and generalization on large-scale, high-frequency data. A relatively low learning rate combined with a large number of boosting iterations allows the model to learn fine-grained patterns in auction dynamics, while the number of leaves controls tree complexity to capture nonlinear feature interactions without excessive overfitting. Early stopping based on validation MAE is applied to automatically select the optimal number of trees.

VI. Evaluation

1) Ridge Regression Performance: The Ridge regression model achieved a mean absolute error (MAE) of 6.31 and a root mean square error (RMSE) of 9.29 on the validation set. We can not surprise with this result that a linear model demonstrated moderate performance when applied to high-frequency auction data, with its RMSE significantly exceeding the MAE. This means model cannot capture extreme price fluctuations—volatility

TABLE I
Ridge Regression Performance

Metric	Value
MAE	6.310143
RMSE	9.286936

2) Ridge Regression Interpretation: Ridge Regression shows that ask and bid prices have the biggest impact, with opposite effects. This means the model simply uses the current price range to predict the auction outcome.

The spread coefficient is negative, showing that predictions are worse when liquidity is low. Imbalance and volume features barely matter in the model—not because they're unimportant, but because Ridge Regression can't handle their nonlinear effects.

In summary, Ridge Regression works as a simple baseline but struggles with complex market patterns.

TABLE II
Ridge Regression Coefficients

Feature	Coefficient	Absolute Coefficient
ask_price	47.812679	47.812679
bid_price	−35.748718	35.748718
wap	−15.441638	15.441638
spread	−9.742468	9.742468
reference_price	3.391361	3.391361
imbalance_pressure	0.179448	0.179448
matched_size	0.010253	0.010253
imbalance_size	0.009611	0.009611

3) Randon Forest Performance: The results reveal that linear price relationships are the primary drivers of short-term auction outcomes, while nonlinear interactions contribute only marginally. Ridge Regression provides a strong baseline by effectively capturing these linear effects (MAE: 6.31). Random Forest, despite its ability to model nonlinear patterns, achieves only slightly worse performance (MAE: 6.45), which indicates that threshold effects and feature interactions are not strongly predictive in this context. These findings suggest that the additional complexity of tree-based models does not translate into improved accuracy for this dataset.

TABLE III
Random Forest Performance

Metric	Value
MAE	6.448341
RMSE	9.848583

4) Randon Forest Interpretation: Feature importance analysis shows that Random Forest relies heavily on price variables. The top three features are weighted average price (0.190), ask price (0.187), and spread (0.169) and all of them account for over 54% of total importance. These features define the current price level and trading range.

Compared to Ridge Regression, the bid-ask spread becomes more important in Random Forest. This shows the model can capture nonlinear liquidity effects, such as threshold behaviors under different market conditions.

Imbalance features contribute moderately (imbalance_pressure: 0.083, imbalance_size: 0.043), suggesting they contain useful but context-dependent signals. The momentum features (matched_size_diff and imbalance_size_diff) have

negligible importance (less than 0.005 each), indicating that short-term fluctuations are averaged out by the ensemble.

Overall, Random Forest still relies primarily on price information, but captures nonlinear patterns better than Ridge Regression.

TABLE IV
RANDOM FOREST FEATURE IMPORTANCE (SORTED BY IMPORTANCE)

Feature	Importance
wap	0.189849
ask_price	0.187088
spread	0.168873
bid_price	0.114010
reference_price	0.108805
matched_size	0.096946
imbalance_pressure	0.083283
imbalance_size	0.042591
matched_size_diff	0.004329
imbalance_size_diff	0.004227

5) LightGBM Model Performance: LightGBM achieves the best performance with MAE of 5.26 and RMSE of 7.63, which outperform both Ridge Regression (MAE: 6.31) and Random Forest (MAE: 6.45). This represents a 16.6% improvement over the linear baseline, demonstrating that gradient boosting effectively captures complex patterns that simpler models miss.

TABLE V
LIGHTGBM PERFORMANCE

Metric	Value
MAE	5.260126
RMSE	7.630974

6) LightGBM Model Performance:Feature Importance Analysis:

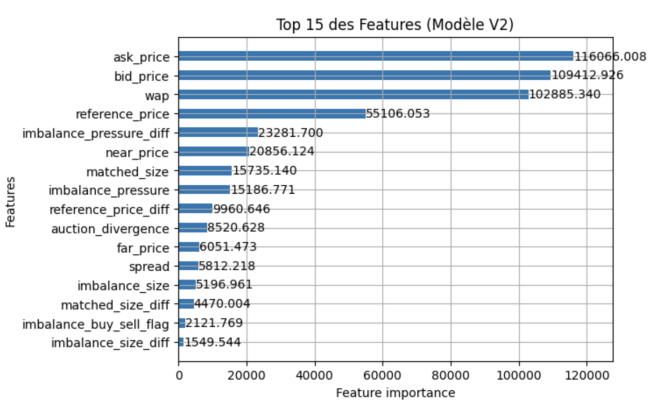

Fig. 1. Feature importance for LightGBM

The feature importance results indicate that price-level variables dominate the predictive performance of the model. The best bid price, best ask price, and weighted average price (WAP) are the most influential features, whch highlight the role of instantaneous price formation in driving short-horizon price movements during the closing auction. The reference price also shows strong importance, confirming its relevance within the auction mechanism.

Beyond static price information, dynamic imbalance-related features contribute meaningfully to the model. In particular, changes in imbalance pressure and reference price differences rank among the most important non-price features, suggesting that rapid shifts in supply–demand pressure provide incremental predictive power. Matched size and imbalance pressure further reinforce the role of order flow intensity in price formation.

In contrast, raw imbalance size and direction alone have lower importance compared to their transformed or differenced counterparts, indicating that the model benefits more from capturing the evolution of imbalance rather than its instantaneous magnitude. Overall, the importance ranking confirms that combining price levels with engineered microstructure and temporal features improves the model's ability to capture closing auction dynamics.

VII. CONCLUSION AND FUTURE WORK

This project demonstrates the feasibility of forecasting stock closing prices using data-driven approaches. Future work may include incorporating macroeconomic indicators and sentiment analysis.

ACKNOWLEDGMENTS

We acknowledge Optiver for providing the dataset and organizing the "Trading at the Close" competition, and Kaggle for hosting the challenge and facilitating access to the data.

REFERENCES

[1] Kaggle. "Optiver – Trading at the Close" dataset. Available at: https://www.kaggle.com/competitions/optiver-trading-at-the-close, 2023.

[2] Rob J. Hyndman and George Athanasopoulos. *Forecasting: Principles and Practice*, 3rd edition, 2021. Available at: https://otexts.com/fpp3/.

[3] Troy J. Strader, John J. Rozycki, Thomas H. Root, and Yu-Hsiang John Huang. Machine learning stock market prediction studies: review and research directions. *Journal of International Technology and Information Management*, 28(4):63–83, 2020.

[4] Kyoung-jae Kim. Financial time series forecasting using support vector machines. *Neurocomputing*, 55(1-2):307–319, 2003. Accessed via Elsevier.

[5] Michel Ballings, Dirk Van den Poel, Nathalie Hespeels, Ruben Gryp. Evaluating multiple classifiers for stock price direction prediction. *Expert Systems with Applications*, 42(20):7046–7056, 2015. Accessed via Elsevier.